

(R)

Statistics/Data Analysis

User: Jakub Murcek
Project: Master Thesis

none	14,190	79.46	79.46
ai_integration	2,354	13.18	92.65
applied_ai	1,307	7.32	99.97
core_ai	6	0.03	100.00
Total	17,857	100.00	

```

1 .
2 . * Procentuální rozdělení (pro text diplomky)
3 . count if has_ai == 1
   3,477

4 . local n_ai = r(N)

5 . count
   17,857

6 . local n_total = r(N)

7 . display _n "Podíl pozic s AI pozadavky: " %5.2f (`n_ai'/`n_total')*100 "%

Podíl pozic s AI pozadavky: 19.47%

```

```

8 .
9 . * --- 4.2 Požadované vzdělání ---
10 . display _n "--- 4.2 Distribuce vzdelavacich pozadavku ---"

--- 4.2 Distribuce vzdelavacich pozadavku ---

```

```
11 . tab edu_cat, missing
```

Pozadovane vzdelani (kategorick e)	Freq.	Percent	Cum.
associate	356	1.99	1.99
bachelor	10,357	58.00	59.99
high school	2	0.01	60.00
highschool	409	2.29	62.29
master	352	1.97	64.27
missing	6,368	35.66	99.93
phd	13	0.07	100.00
Total	17,857	100.00	

```
12 . tab edu_cat if edu_cat > 0
```

Pozadovane vzdelani (kategorick e)	Freq.	Percent	Cum.
associate	356	1.99	1.99
bachelor	10,357	58.00	59.99
high school	2	0.01	60.00
highschool	409	2.29	62.29
master	352	1.97	64.27
missing	6,368	35.66	99.93
phd	13	0.07	100.00
Total	17,857	100.00	

```

13 .
14 . * Porovnání vzdělání podle AI tier
15 . display _n "Vzdelani x AI tier (kontingencni tabulka):"

```

Vzdelani x AI tier (kontingencni tabulka):

```

16 . tab edu_cat ai_tier_num if edu_cat > 0, chi2 column

```

Key
<i>frequency</i>
<i>column percentage</i>

Pozadovane vzdelani (kategorick e)	ai_tier_num				Total
	ai_integr	applied_a	core_ai	none	
associate	20 0.85	9 0.69	0 0.00	327 2.30	356 1.99
bachelor	1,214 51.57	702 53.71	1 16.67	8,440 59.48	10,357 58.00
high school	0 0.00	0 0.00	0 0.00	2 0.01	2 0.01
highschool	28 1.19	20 1.53	0 0.00	361 2.54	409 2.29
master	26 1.10	39 2.98	2 33.33	285 2.01	352 1.97
missing	1,066 45.28	532 40.70	2 33.33	4,768 33.60	6,368 35.66
phd	0 0.00	5 0.38	1 16.67	7 0.05	13 0.07
Total	2,354 100.00	1,307 100.00	6 100.00	14,190 100.00	17,857 100.00

Pearson chi2(18) = 462.1521 Pr = 0.000

```

17 .
18 . * --- 4.3 Požadované zkušenosti ---
19 . display _n "--- 4.3 Distribuce pozadavku na zkusenosti ---"

```

--- 4.3 Distribuce pozadavku na zkusenosti ---

```

20 . summarize experience_min_llm, detail

```

Min. pozadovane roky zkusenosti					
	Percentiles	Smallest			
1%	0	0			
5%	.5	0			
10%	1	0	Obs	15,262	
25%	3	0	Sum of Wgt.	15,262	
50%	5		Mean	4.520393	
		Largest	Std. Dev.	2.62528	
75%	5	20			
90%	8	25	Variance	6.892093	
95%	10	25	Skewness	.9727604	
99%	12	35	Kurtosis	6.38147	

21 .
 22 . tab exp_category, missing

Kategorie seniority	Freq.	Percent	Cum.
Entry (0)	3,348	18.75	18.75
Junior (1-2)	2,460	13.78	32.53
Mid (3-5)	8,411	47.10	79.63
Senior (6-10)	3,419	19.15	98.77
Expert (10+)	219	1.23	100.00
Total	17,857	100.00	

23 . tab exp_category ai_tier_num, chi2 column

Key
<i>frequency</i> <i>column percentage</i>

Kategorie seniority	ai_integr	ai_tier_num applied_a	core_ai	none	Total
Entry (0)	341 14.49	222 16.99	1 16.67	2,784 19.62	3,348 18.75
Junior (1-2)	276 11.72	153 11.71	1 16.67	2,030 14.31	2,460 13.78
Mid (3-5)	1,238 52.59	618 47.28	2 33.33	6,553 46.18	8,411 47.10
Senior (6-10)	472 20.05	289 22.11	2 33.33	2,656 18.72	3,419 19.15
Expert (10+)	27 1.15	25 1.91	0 0.00	167 1.18	219 1.23
Total	2,354 100.00	1,307 100.00	6 100.00	14,190 100.00	17,857 100.00

Pearson chi2(12) = **77.2871** Pr = **0.000**

24 .
 25 . * --- 4.4 Platy ---
 26 . display_n "--- 4.4 Distribuce platu ---"

--- 4.4 Distribuce platu ---

27 . summarize salary_mid, detail

Rocni plat - stredni hodnota (USD)				
Percentiles	Smallest			
1%	56538	20000		
5%	70937	30000		
10%	79863	30000	Obs	13,743
25%	95000	32640	Sum of Wgt.	13,743
50%	118480		Mean	125240.4
		Largest	Std. Dev.	42210
75%	148750	445000		
90%	180000	445000	Variance	1.78e+09
95%	200000	445000	Skewness	1.370318
99%	245000	450000	Kurtosis	7.794587

```

28 .
29 . * Platy podle AI tier - DŮLEŽITĚ pro argument o "AI premium"
30 . display _n "Plat podle AI tier:"

```

Plat podle AI tier:

```

31 . tabstat salary_mid, by(desc_tier_llm) statistics(count mean sd min p25 p50 p75 max)

```

Summary for variables: salary_mid
by categories of: desc_tier_llm

desc_tier_llm	N	mean	sd	min	p25	p50	p75	max
ai_integration	1869	141477.4	46393.67	36500	107987	135000	169600	410000
applied_ai	1041	151004.5	48527	46532	116373	149500	180000	445000
core_ai	4	168045.8	55785.28	100558	129279	167937.5	206812.5	235750
none	10829	119945.5	39048.28	20000	92500	114067	140000	450000
Total	13743	125240.4	42210	20000	95000	118480	148750	450000

```

32 .
33 . * --- 4.5 Sektory a industrie ---
34 . display _n "--- 4.5 Top 10 sektoru ---"

```

--- 4.5 Top 10 sektoru ---

```

35 . tab sector if sector != "Unknown", sort

```

sector	Freq.	Percent	Cum.
Information Technology	6,074	42.32	42.32
Manufacturing	1,219	8.49	50.81
Financial Services	1,081	7.53	58.34
Aerospace & Defense	981	6.83	65.18
Management & Consulting	707	4.93	70.10
Healthcare	692	4.82	74.93
Media & Communication	546	3.80	78.73
Retail & Wholesale	405	2.82	81.55
Education	375	2.61	84.16
Insurance	321	2.24	86.40
Human Resources & Staffing	294	2.05	88.45
Government & Public Administration	271	1.89	90.34
Pharmaceutical & Biotechnology	229	1.60	91.93
Energy, Mining & Utilities	184	1.28	93.21
Telecommunications	175	1.22	94.43
Construction, Repair & Maintenance Se..	171	1.19	95.62
Transportation & Logistics	143	1.00	96.62
Arts, Entertainment & Recreation	103	0.72	97.34
Real Estate	94	0.65	97.99
Nonprofit & NGO	76	0.53	98.52
Personal Consumer Services	56	0.39	98.91
Restaurants & Food Service	48	0.33	99.25
Legal	44	0.31	99.55
Hotels & Travel Accommodation	33	0.23	99.78
Agriculture	30	0.21	99.99
Media	1	0.01	100.00
Total	14,353	100.00	

```
36 .
37 . display _n "--- 4.6 Remote prace ---"
```

```
--- 4.6 Remote prace ---
```

```
38 . tab is_remote
```

Moznost remote prace (1=ano)	Freq.	Percent	Cum.
0	17,857	100.00	100.00
Total	17,857	100.00	

```
39 . tab is_remote has_ai, chi2 row
```

Key
<i>frequency</i> <i>row percentage</i>

Moznost remote prace (1=ano)	has_ai		Total
	0	1	
0	14,380 80.53	3,477 19.47	17,857 100.00
Total	14,380 80.53	3,477 19.47	17,857 100.00

```
40 .
41 .
42 . * =====
43 . * 5. ANALYTICKÉ TESTY - HYPOTÉZY
44 . * =====
45 . * Statistické testy pro ověření hypotéz diplomové práce
46 .
47 . display _n "=====
```

```
=====
```

```
48 . display "5. STATISTICKE TESTY"
5. STATISTICKE TESTY
```

```
49 . display "=====
```

```
=====
```

```
50 .
51 . * --- 5.1 T-test: Liší se platy AI vs non-AI pozic? ---
52 . * H0: Průměrný plat AI pozic = průměrný plat non-AI pozic
53 . * H1: Průměrné platy se liší
54 .
55 . display _n "--- 5.1 T-test: Plat AI vs non-AI pozic ---"
```

```
--- 5.1 T-test: Plat AI vs non-AI pozic ---
```

```
56 . ttest salary_mid, by(has_ai)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
0	10,988	120345.9	375.8993	39403.15	119609.1	121082.8
1	2,755	144761.4	897.5534	47110.87	143001.5	146521.4
combined	13,743	125240.4	360.0597	42210	124534.6	125946.2
diff		-24415.49	874.948		-26130.5	-22700.47

```
diff = mean(0) - mean(1)                                t = -27.9051
Ho: diff = 0                                             degrees of freedom = 13741
```

```
Ha: diff < 0                                Ha: diff != 0                                Ha: diff > 0
Pr(T < t) = 0.0000                        Pr(|T| > |t|) = 0.0000                        Pr(T > t) = 1.0000
```

```
57 .
```

```
58 . * Efekt size (Cohenovo d) - důležité pro interpretaci praktické významnosti
```

```
59 . quietly summarize salary_mid if has_ai == 0
```

```
60 . local mean_no_ai = r(mean)
```

```
61 . local sd_no_ai = r(sd)
```

```
62 . local n_no_ai = r(N)
```

```
63 .
```

```
64 . quietly summarize salary_mid if has_ai == 1
```

```
65 . local mean_ai = r(mean)
```

```
66 . local sd_ai = r(sd)
```

```
67 . local n_ai = r(N)
```

```
68 .
```

```
69 . * Pooled standard deviation
```

```
70 . local sd_pooled = sqrt(((`n_no_ai'-1)*`sd_no_ai'^2 + (`n_ai'-1)*`sd_ai'^2) / (`n_no_ai'+`n_ai'))
```

```
71 . local cohens_d = (`mean_ai' - `mean_no_ai') / `sd_pooled'
```

```
72 .
```

```
73 . display _n "Cohenovo d (effect size): " %5.3f `cohens_d'
```

Cohenovo d (effect size): 0.595

```
74 . display "Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt"
Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt
```

```
75 .
```

```
76 . * --- 5.2 ANOVA: Liší se platy mezi AI tiers? ---
```

```
77 . * Testuje rozdíly mezi none, ai_integration, ai_focused
```

```
78 .
```

```
79 . display _n "--- 5.2 ANOVA: Plat podle AI tier ---"
```

--- 5.2 ANOVA: Plat podle AI tier ---

```
80 . oneway salary_mid ai_tier_num, tabulate bonferroni
```

ai_tier_num	Summary of Rocni plat - stredni hodnota (USD)		
	Mean	Std. Dev.	Freq.
ai_integr	141477.39	46393.668	1,869
applied_a	151004.53	48526.999	1,041
core_ai	168045.75	55785.282	4
none	119945.51	39048.279	10,829
Total	125240.41	42209.997	13,743

Source	Analysis of Variance			F	Prob > F
	SS	df	MS		
Between groups	1.4947e+12	3	4.9823e+11	297.75	0.0000
Within groups	2.2989e+13	13739	1.6733e+09		
Total	2.4484e+13	13742	1.7817e+09		

Bartlett's test for equal variances: $\chi^2(3) = 172.4292$ Prob> $\chi^2 = 0.000$

Comparison of Rocni plat - stredni hodnota (USD) by ai_tier_num
(Bonferroni)

Row Mean- Col Mean	ai_integ	applied_	core_ai
applied_	9527.14 0.000		
core_ai	26568.4 1.000	17041.2 1.000	
none	-21531.9 0.000	-31059 0.000	-48100.2 0.112

```
81 .
82 . * --- 5.3 Chi-square: Vzdelání a AI požadavky ---
83 . * Jsou AI pozice náročnější na vzdělání?
84 .
85 . display _n "--- 5.3 Chi-square: Vzdelani x AI tier ---"
```

--- 5.3 Chi-square: Vzdelani x AI tier ---

```
86 . tab edu_cat has_ai if edu_cat > 0, chi2 expected
```

Key
<i>frequency</i> <i>expected frequency</i>

Pozadovane vzdelani (kategorick e)	has_ai		Total
	0	1	
associate	328 286.7	28 69.3	356 356.0
bachelor	8,543 8,340.4	1,814 2,016.6	10,357 10,357.0
high school	2 1.6	0 0.4	2 2.0
highschool	363 329.4	46 79.6	409 409.0
master	286	66	352

	283.5	68.5	352.0
missing	4,851 5,128.1	1,517 1,239.9	6,368 6,368.0
phd	7 10.5	6 2.5	13 13.0
Total	14,380 14,380.0	3,477 3,477.0	17,857 17,857.0

Pearson $\chi^2(6) = 156.8968$ Pr = 0.000

```
87 .
88 . * --- 5.4 Chi-square: Zkušenosti a AI požadavky ---
89 . display _n "--- 5.4 Chi-square: Zkusenosti x AI tier ---"
```

--- 5.4 Chi-square: Zkusenosti x AI tier ---

```
90 . tab exp_category has_ai, chi2 expected
```

Key
<i>frequency</i> <i>expected frequency</i>

Kategorie seniority	has_ai		Total
	0	1	
Entry (0)	2,809 2,696.1	539 651.9	3,348 3,348.0
Junior (1-2)	2,052 1,981.0	408 479.0	2,460 2,460.0
Mid (3-5)	6,655 6,773.3	1,756 1,637.7	8,411 8,411.0
Senior (6-10)	2,693 2,753.3	726 665.7	3,419 3,419.0
Expert (10+)	171 176.4	48 42.6	219 219.0
Total	14,380 14,380.0	3,477 3,477.0	17,857 17,857.0

Pearson $\chi^2(4) = 55.5656$ Pr = 0.000

```
91 .
92 . * --- 5.5 Mann-Whitney U test (neparametrický) ---
93 . * Pro případ, že platy nemají normální rozdělení
94 .
95 . display _n "--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---"
```

--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---

```
96 . ranksum salary_mid, by(has_ai)
```

Two-sample Wilcoxon rank-sum (Mann-Whitney) test

has_ai	obs	rank sum	expected
0	10988	70557704	75509536
1	2755	23884192	18932360
combined	13743	94441896	94441896


```

unadjusted variance    3.467e+10
adjustment for ties    -595872.38
-----
adjusted variance      3.467e+10

```

```

Ho: salary~d(has_ai==0) = salary~d(has_ai==1)
      z = -26.594
      Prob > |z| = 0.0000

```

```

97 .
98 .
99 . * =====
100 . * 6. REGRESNÍ ANALÝZA
101 . * =====
102 . * Modelování faktorů ovlivňujících plat a AI požadavky
103 .
104 . display _n "=====

```

```
=====
```

```

105 . display "6. REGRESNI ANALYZA"
6. REGRESNI ANALYZA

```

```

106 . display "=====
=====

```

```

107 .
108 . * --- 6.1 OLS regrese: Co ovlivňuje plat? ---
109 . * Závislá proměnná: salary_mid
110 . * Nezávislé: has_ai, edu_level, experience_min_llm, is_remote
111 .
112 . display _n "--- 6.1 OLS regrese: Determinanty platu ---"

```

--- 6.1 OLS regrese: Determinanty platu ---

```

113 .
114 . regress salary_mid has_ai i.edu_cat experience_min_llm is_remote
note: is_remote omitted because of collinearity

```

Source	SS	df	MS	Number of obs	=	11,899
Model	4.0788e+12	8	5.0985e+11	F(8, 11890)	=	370.79
Residual	1.6349e+13	11,890	1.3750e+09	Prob > F	=	0.0000
				R-squared	=	0.1997
				Adj R-squared	=	0.1991
Total	2.0428e+13	11,898	1.7169e+09	Root MSE	=	37082

salary_mid	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
has_ai	21441.24	848.7631	25.26	0.000	19777.52	23104.95
edu_cat						
bachelor	11283.6	2343.494	4.81	0.000	6689.964	15877.23
high school	-21236.24	37153.58	-0.57	0.568	-94063.33	51590.85
highschool	6134.231	3109.041	1.97	0.049	40.00268	12228.46
master	17723.23	3332.786	5.32	0.000	11190.43	24256.04
missing	19700.28	2396.148	8.22	0.000	15003.44	24397.12
phd	36161.52	13326.74	2.71	0.007	10038.93	62284.1
experience_min_llm	5676.079	132.5348	42.83	0.000	5416.289	5935.869
is_remote	0	(omitted)				
_cons	82449.16	2329.612	35.39	0.000	77882.74	87015.58

```

115 .
116 . * Robustní standardní chyby (heteroskedasticita)
117 . display _n "--- 6.1b OLS s robustními standardními chybami ---"

```

--- 6.1b OLS s robustními standardními chybami ---

```

118 . regress salary_mid has_ai i.edu_cat experience_min_llm is_remote, vce(robust)
note: is_remote omitted because of collinearity

```

```

Linear regression              Number of obs      =      11,899
                              F(7, 11890)          =      .
                              Prob > F              =      .
                              R-squared              =      0.1997
                              Root MSE           =      37082

```

salary_mid	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
has_ai	21441.24	932.6863	22.99	0.000	19613.02	23269.46
edu_cat						
bachelor	11283.6	2107.281	5.35	0.000	7152.98	15414.21
high school	-21236.24	2067.935	-10.27	0.000	-25289.73	-17182.75
highschool	6134.231	2649.912	2.31	0.021	939.971	11328.49
master	17723.23	3078.04	5.76	0.000	11689.77	23756.69
missing	19700.28	2208.873	8.92	0.000	15370.53	24030.03
phd	36161.52	14946.56	2.42	0.016	6863.806	65459.23
experience_min_llm	5676.079	161.0389	35.25	0.000	5360.416	5991.741
is_remote	0	(omitted)				
_cons	82449.16	2093.398	39.39	0.000	78345.76	86552.56

```

119 .
120 . * --- 6.2 Rozšířený model s interakcemi ---
121 . * AI × vzdělání interakce - má AI větší vliv u vyššího vzdělání?
122 .
123 . display _n "--- 6.2 Model s interakcemi ---"

```

--- 6.2 Model s interakcemi ---

```

124 . regress salary_mid c.has_ai##i.edu_cat experience_min_llm is_remote, vce(robust)
note: 3.edu_cat#c.has_ai omitted because of collinearity
note: is_remote omitted because of collinearity

```

```

Linear regression              Number of obs      =      11,899
                              F(12, 11885)         =      .
                              Prob > F              =      .
                              R-squared              =      0.2020
                              Root MSE           =      37035

```

salary_mid	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
has_ai	18954.81	9463.682	2.00	0.045	404.4479	37505.18
edu_cat						
bachelor	11667.26	2124.869	5.49	0.000	7502.166	15832.35
high school	-21446.66	2082.054	-10.30	0.000	-25527.83	-17365.5
highschool	7887.785	2696.078	2.93	0.003	2603.032	13172.54
master	15997.48	3082.677	5.19	0.000	9954.93	22040.03
missing	17838.6	2250.527	7.93	0.000	13427.2	22250
phd	33998.26	9670.165	3.52	0.000	15043.15	52953.37
edu_cat#c.has_ai						
bachelor	-939.3974	9527.004	-0.10	0.921	-19613.88	17735.09
high school	0	(omitted)				
highschool	-13850.99	10913.94	-1.27	0.204	-35244.09	7542.113
master	9453.47	11745.45	0.80	0.421	-13569.53	32476.47
missing	8319.134	9620.865	0.86	0.387	-10539.33	27177.6

phd	6335.796	31040.25	0.20	0.838	-54508.18	67179.77
experience_min_llm	5680.494	161.0209	35.28	0.000	5364.867	5996.121
is_remote	0	(omitted)				
_cons	82655.17	2107.169	39.23	0.000	78524.77	86785.56

```

125 .
126 . * --- 6.3 Logistická regrese: Co predikuje AI požadavky? ---
127 . * Které charakteristiky pozice souvisí s AI požadavky?
128 .
129 . display _n "--- 6.3 Logisticka regrese: Prediktory AI pozadavku ---"

```

--- 6.3 Logisticka regrese: Prediktory AI pozadavku ---

```
130 . logit has_ai i.edu_cat experience_min_llm is_remote, or
```

```
note: 3.edu_cat != 0 predicts failure perfectly
      3.edu_cat dropped and 2 obs not used
```

```
note: is_remote omitted because of collinearity
```

```
Iteration 0: log likelihood = -7640.298
```

```
Iteration 1: log likelihood = -7540.7908
```

```
Iteration 2: log likelihood = -7538.3973
```

```
Iteration 3: log likelihood = -7538.385
```

```
Iteration 4: log likelihood = -7538.385
```

Logistic regression	Number of obs	=	15,260
	LR chi2(6)	=	203.83
	Prob > chi2	=	0.0000
Log likelihood = -7538.385	Pseudo R2	=	0.0133

has_ai	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]
edu_cat					
bachelor	2.413172	.4982747	4.27	0.000	1.610019 3.616974
high school	1	(empty)			
highschool	1.447353	.3775625	1.42	0.156	.8680134 2.413362
master	2.836938	.7158161	4.13	0.000	1.730114 4.651842
missing	3.8925	.8073536	6.55	0.000	2.592261 5.84492
phd	8.393511	5.901799	3.03	0.002	2.115589 33.30091
experience_min_llm	1.026246	.0079106	3.36	0.001	1.010858 1.041869
is_remote	1	(omitted)			
_cons	.0792353	.016307	-12.32	0.000	.0529343 .1186042

Note: **_cons** estimates baseline odds.

```

131 .
132 . * Marginální efekty (interpretovatelné jako procentní změny)
133 . display _n "--- 6.3b Marginalni efekty ---"

```

--- 6.3b Marginalni efekty ---

```
134 . margins, dydx(*) atmeans
```

Conditional marginal effects	Number of obs	=	15,260
Model VCE : OIM			

```
Expression : Pr(has_ai), predict()
```

```
dy/dx w.r.t. : 2.edu_cat 3.edu_cat 4.edu_cat 5.edu_cat 6.edu_cat 7.edu_cat experience_min_llm i
```

```
at : 1.edu_cat = .0215596 (mean)
```

```
2.edu_cat = .6127785 (mean)
```

```
4.edu_cat = .0249672 (mean)
```

```
5.edu_cat = .0189384 (mean)
```

```
6.edu_cat = .3211664 (mean)
```

```
7.edu_cat = .0005898 (mean)
```

```
experience~m = 4.520724 (mean)
```

```
is_remote = 0 (mean)
```

	dy/dx	Delta-method Std. Err.	z	P> z	[95% Conf. Interval]	
edu_cat						
bachelor	.0951381	.0158743	5.99	0.000	.0640251	.1262512
high school	.	(not estimable)				
highschool	.0324121	.0224549	1.44	0.149	-.0115986	.0764228
master	.1199405	.0283431	4.23	0.000	.064389	.175492
missing	.1756757	.0166229	10.57	0.000	.1430954	.2082559
phd	.3460253	.1653303	2.09	0.036	.0219839	.6700667
experience_min_llm	.0040822	.0012136	3.36	0.001	.0017035	.0064609
is_remote	0	(omitted)				

Note: dy/dx for factor levels is the discrete change from the base level.

```

135 .
136 .
137 . * =====
138 . * 7. ANALÝZA HARD SKILLS
139 . * =====
140 . * Rozbor nejčastějších technických dovedností
141 .
142 . display _n "=====
=====

143 . display "7. ANALÝZA HARD SKILLS"
7. ANALÝZA HARD SKILLS

144 . display "=====
=====

145 .
146 . * Poznámka: hardskills je textový sloupec s čárkami oddělenými skills
147 . * Pro detailní analýzu je lepší použít Python k vytvoření dummy proměnných
148 . * Zde ukážeme základní exploraci
149 .
150 . * Počet unique skills na pozici (aproximace)
151 . gen skill_count = 1 + length(hardskills) - length(subinstr(hardskills, ",", "", .))

152 . replace skill_count = 0 if hardskills == ""
(110 real changes made)

153 . label variable skill_count "Pocet pozadovanych hard skills"

154 .
155 . display _n "--- 7.1 Pocet skills na pozici ---"

--- 7.1 Pocet skills na pozici ---

156 . summarize skill_count, detail

```

Pocet pozadovanych hard skills

Percentiles		Smallest		
1%	1	0		
5%	3	0		
10%	5	0	Obs	17,857
25%	9	0	Sum of Wgt.	17,857
50%	15		Mean	16.78658
		Largest	Std. Dev.	10.30584
75%	23	72		
90%	31	72	Variance	106.2103
95%	36	80	Skewness	.9649615
99%	47	80	Kurtosis	4.325341

```
157 . tabstat skill_count, by(desc_tier_llm) statistics(count mean sd min max)
```

Summary for variables: skill_count
by categories of: desc_tier_llm

desc_tier_llm	N	mean	sd	min	max
ai_integration	2354	19.46432	10.3817	0	68
applied_ai	1307	20.62663	10.79929	1	80
core_ai	6	11.16667	6.431692	1	20
none	14190	15.99105	10.09171	0	80
Total	17857	16.78658	10.30584	0	80

```
158 .
159 . * T-test: Vyžadují AI pozice více skills?
160 . display _n "--- 7.2 T-test: Pocet skills AI vs non-AI ---"
```

--- 7.2 T-test: Pocet skills AI vs non-AI ---

```
161 . ttest skill_count, by(has_ai)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
0	14,380	16.04652	.0842893	10.10769	15.88131	16.21174
1	3,477	19.84728	.1789442	10.55164	19.49644	20.19813
combined	17,857	16.78658	.0771222	10.30584	16.63542	16.93775
diff		-3.800759	.19268		-4.178431	-3.423088

diff = mean(0) - mean(1) t = -19.7258
Ho: diff = 0 degrees of freedom = 17855

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.0000 Pr(|T| > |t|) = 0.0000 Pr(T > t) = 1.0000

```
162 .
163 .
164 . * =====
165 . * 8. EXPORTY PRO TABULKY A GRAFY
166 . * =====
167 . * Příprava dat pro publikovatelné tabulky
168 .
169 . display _n "====="
```

=====

```
170 . display "8. EXPORTY"
170 . 8. EXPORTY
```

```
171 . display "=====
171 . ====="
```

```
172 .
173 . * --- 8.1 Summary statistics ---
174 . display _n "--- 8.1 Summary statistics pro tabulky ---"
```

--- 8.1 Summary statistics pro tabulky ---

```
175 . summarize salary_mid experience_min_llm edu_cat skill_count has_ai is_remote
```

Variable	Obs	Mean	Std. Dev.	Min	Max
salary_mid	13,743	125240.4	42210	20000	450000
experience~m	15,262	4.520393	2.62528	0	35
edu_cat	17,857	3.515204	1.928059	1	7
skill_count	17,857	16.78658	10.30584	0	80
has_ai	17,857	.1947136	.3959911	0	1
is_remote	17,857	0	0	0	0

```
176 .
177 . * Export do CSV pro další zpracování (grafy v Excelu/R/Python)
178 . export delimited desc_tier_llm salary_mid experience_min_llm edulevel_llm state sector using
    (note: file ./output/summary_for_charts.csv not found)
    file ./output/summary_for_charts.csv saved

179 .
180 .
181 . * =====
182 . * 9. VIZUALIZACE
183 . * =====
184 . * Základní grafy pro diplomovou práci
185 .
186 . display _n "=====
    =====

187 . display "9. VIZUALIZACE"
9. VIZUALIZACE

188 . display "=====
    =====

189 .
190 . * --- 9.1 Sloupcový graf: AI tier distribuce ---
191 . graph bar (count), over(desc_tier_llm) ///
    > title("Distribuce AI požadavku v IT pozicích") ///
    > ytitle("Pocet pozic") ///
    > bar(1, color(navy))

192 . graph export "$outdir/ai_tier_distribution.png", replace width(1200)
    (note: file ./output/ai_tier_distribution.png not found)
    (file ./output/ai_tier_distribution.png written in PNG format)

193 .
194 . * --- 9.2 Box plot: Platy podle AI tier ---
195 . graph box salary_mid, over(desc_tier_llm) ///
    > title("Rozložení platu podle AI požadavku") ///
    > ytitle("Rocni plat (USD)")

196 . graph export "$outdir/salary_by_ai_tier.png", replace width(1200)
    (note: file ./output/salary_by_ai_tier.png not found)
    (file ./output/salary_by_ai_tier.png written in PNG format)

197 .
198 . * --- 9.3 Histogram: Požadované zkušenosti ---
199 . histogram experience_min_llm if experience_min_llm < 15, ///
    > by(has_ai, title("Požadovane zkušenosti") note("")) ///
    > xtitle("Roky zkušenosti") percent bin(15)
```

```

200 . graph export "$outdir/experience_histogram.png", replace width(1200)
    (note: file ./output/experience_histogram.png not found)
    (file ./output/experience_histogram.png written in PNG format)

201 .
202 . * --- 9.4 Vzdělání podle AI tier ---
203 . graph bar (count), over(edu_cat) over(has_ai) ///
    > title("Vzdelavaci pozadavky: AI vs non-AI pozice") ///
    > ytitle("Pocet pozic") ///
    > legend(label(1 "non-AI") label(2 "AI pozice"))

204 . graph export "$outdir/education_by_ai.png", replace width(1200)
    (note: file ./output/education_by_ai.png not found)
    (file ./output/education_by_ai.png written in PNG format)

205 .
206 .
207 . * =====
208 . * 10. ZÁVĚR A ULOŽENÍ
209 . * =====
210 .
211 . display _n "=====
    =====

212 . display "ANALYZA DOKONCENA"
    ANALYZA DOKONCENA

213 . display "=====
    =====

214 . display "Vystupy ulozeny do: $outdir"
    Vystupy ulozeny do: ./output

215 . display "Log soubor: $outdir/ai_skills_analysis.log"
    Log soubor: ./output/ai_skills_analysis.log

216 .
217 . * Ulož zpracovaný dataset pro další práci
218 . save "$outdir/ai_skills_processed.dta", replace
    (note: file ./output/ai_skills_processed.dta not found)
    file ./output/ai_skills_processed.dta saved

219 .
220 . * Zavři log
221 . log close
    name: <unnamed>
    log: C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output\ai_skills_a
    log type: text
    closed on: 24 Jan 2026, 12:02:54

222 .
223 . * =====
224 . * KONEC DO-FILU
225 . * =====
226 .
    end of do-file

227 . do "C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\ai_skills_analysis.do"

```

```

228 . *****
229 . * AI SKILLS IN IT JOB POSTINGS - STATA ANALYSIS
230 . * =====
231 . * Dataset: us_relevant_ai.csv
232 . * Autor: [Yakub Murcek]
233 . * Datum: Leden 2026
234 . *
235 . * Tento do-file obsahuje kompletní analýzu datasetu IT pracovních inzerátů
236 . * s fokusem na požadavky na AI dovednosti.
237 . *
238 . * TESTOVÁNO PRO: STATA 15.1 (IC/SE verze)
239 . *****
240 .
241 . * =====
242 . * 1. NASTAVENÍ PROSTŘEDÍ
243 . * =====
244 . * Vyčistíme paměť a nastavíme working directory
245 . * Toto je důležité pro reprodukovatelnost - každý běh začíná čistě
246 . * AUTO-SETUP: Pokus o automatické nastavení složky (Windows)
247 . capture {

248 . if _rc {
249 .     display as text "Pozor: Nepodařilo se automaticky nastavit složku. Ujistěte se, že jste v
250 . }

251 . display "Aktuální pracovní složka: " c(pwd)
Aktuální pracovní složka: C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata

252 .
253 . version 15.1

254 . clear all

255 . set more off

256 .
257 . * Nastav cestu k datům - UPRAV PODLE SVÉ STRUKTURY
258 . global datadir "../..data/outputs"

259 . global outdir "./output"

260 .
261 . * Vytvoř output složku pokud neexistuje
262 . capture mkdir "$outdir"

263 .
264 . * Log file - zaznamenává veškerý výstup pro pozdější kontrolu
265 . capture log close

266 . * Generování časového razítka pro unikátní název log souboru
267 . local c_date = c(current_date)

268 . local c_time = c(current_time)

269 . local time_string = substr("`c_date'`c_time'", " ", "_", .)

270 . local time_string = substr("`time_string'", ":", "-", .)

271 . log using "$outdir/ai_skills_analysis_`time_string'.log", replace text
(note: file C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output\ai_skills_a

name: <unnamed>
log: C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output\ai_skills_a
log type: text
opened on: 24 Jan 2026, 12:27:12

```



```

272 .
273 .
274 . * =====
275 . * 2. IMPORT DAT
276 . * =====
277 . * Používáme soubor 'us_relevant_ai_stata.csv'
278 . import delimited "$datadir/us_relevant_ai_stata.csv", delimiter(";") clear varnames(1) encoding(utf8)
    (72 vars, 18,464 obs)

279 .
280 . * Základní kontrola úspěšného importu
281 . describe

```

Contains data

```

    obs:      18,464
    vars:       72
    size:    88,479,488

```

variable name	storage type	display format	value label	variable label
skills	str576	%576s		
skills_ai_det	str98	%98s		
skills_hasai_~t	byte	%8.0g		
desc_tier_llm	str14	%14s		
desc_conf_llm	float	%9.0g		
ai_confidence	float	%9.0g		
desc_ai_llm	str400	%400s		
is_real_ai	byte	%8.0g		
hardskills	str934	%934s		
softskills	str452	%452s		
edu_level_det	str10	%10s		
edulevel_llm	str11	%11s		
experience_mi~m	float	%9.0g		
ai_det_llm_ma~h	byte	%8.0g		
desc_hard_det	str934	%934s		
desc_soft_det	str452	%452s		
desc_hard_llm	byte	%8.0g		
desc_soft_llm	byte	%8.0g		
id	int	%8.0g		
job_title	str131	%131s		
location	str32	%32s		
company_id	long	%12.0g		
company	str81	%81s		
age_in_days	int	%8.0g		
pay_currency	str3	%9s		
pay_period	str7	%9s		
salary_min	float	%9.0g		
salary_mid	float	%9.0g		
salary_max	float	%9.0g		
rating	float	%9.0g		
discover_date	str19	%19s		
job_types	str73	%73s		
remote_work_t~s	str14	%14s		
city	str32	%32s		
state	str24	%24s		
country	str13	%13s		
latitude	float	%9.0g		
longitude	float	%9.0g		
ceo	str64	%64s		
headquarters	str37	%37s		
industry	str40	%40s		
sector	str43	%43s		
revenue	str32	%32s		
size	str23	%23s		
type	str30	%30s		
website	str127	%127s		
year_founded	int	%8.0g		
cluster_archi~s	byte	%8.0g		
cluster_bi____~s	byte	%8.0g		
cluster_backe~t	byte	%8.0g		
cluster_certi~s	byte	%8.0g		

```

cluster_cloud~g byte      %8.0g
cluster_data_~s byte      %8.0g
cluster_data_~g byte      %8.0g
cluster_data_~l byte      %8.0g
cluster_datab~e byte      %8.0g
cluster_devp~s byte      %8.0g
cluster_dynam~b byte      %8.0g
cluster_enter~d byte      %8.0g
cluster_enter~s byte      %8.0g
cluster_front~t byte      %8.0g
cluster_gener~i byte      %8.0g
cluster_legac~e byte      %8.0g
cluster_mobil~p byte      %8.0g
cluster_netwo~g byte      %8.0g
cluster_os___~d byte      %8.0g
cluster_scrip~l byte      %8.0g
cluster_secur~y byte      %8.0g
cluster_syste~g byte      %8.0g
cluster_testi~g byte      %8.0g
cluster_tools~s byte      %8.0g
education_hyb~d str11     %11s

```

Sorted by:

Note: Dataset has changed since last saved.

```

282 . display "Počet pozorování (job postů): " _N
    Počet pozorování (job postů): 18464

```

```

283 .
284 .
285 . * =====
286 . * 3. ČIŠTĚNÍ A PŘÍPRAVA DAT
287 . * =====
288 .
289 . * --- 3.0 Filtrování (podle požadavku uživatele) ---
290 . * Ponecháme pouze pozice s vysokou confidence modelu
291 . keep if desc_conf_llm >= 0.7
    (607 observations deleted)

```

```

292 . display "Počet pozorování po filtrování confidence >= 0.7: " _N
    Počet pozorování po filtrování confidence >= 0.7: 17857

```

```

293 .
294 . * --- 3.1 AI Tier klasifikace ---
295 . * desc_tier_llm obsahuje kategorie: "none", "ai_integration", "applied_ai"
296 . replace desc_tier_llm = "missing" if desc_tier_llm == ""
    (0 real changes made)

```

```

297 . encode desc_tier_llm, generate(ai_tier_num)

```

```

298 .
299 . * --- 3.2 AI Flag (Strict Intersection + Buzzword Filter) ---
300 . * 1. Sloučíme zdroje skills do jednoho retezce pro kontrolu
301 . gen skills_combined = lower(skills_ai_det + " " + desc_ai_llm)

```

```

302 .
303 . * 2. Odstraníme obecné buzzwords (AI, ML, Artificial Intelligence atd.)
304 . * Používáme regex s word boundaries \b aby se smazalo jen "AI" a ne "OpenAI"
305 . gen skills_no_buzz = ustrregexra(skills_combined, "(?i)\b(ai|ml|artificial intelligence|machi

```

```

306 .
307 . * 3. Odstraníme interpunkci (carky, mezery), abychom zaručili ze zbyva neco realneho
308 . gen skills_cleaned = substr(skills_no_buzz, "", "", .)

309 . replace skills_cleaned = substr(skills_cleaned, " ", "", .)
    (17,857 real changes made)

310 . replace skills_cleaned = substr(skills_cleaned, ";", "", .)
    (0 real changes made)

311 .
312 . * 4. Definice has_ai_flag
313 . * Tier != None A ZAROVEN zbyly nejake specificke skills (delka > 1 znaku)
314 . gen has_ai_flag = ((desc_tier_llm != "none" & desc_tier_llm != "missing") & length(skills_cleaned) > 1)

315 .
316 . label variable has_ai_flag "AI Job (Tier + Valid Skills, no buzzwords)"

317 .
318 . * Kompatibilita pro starší skripty (volitelné)
319 . gen has_ai = has_ai_flag

320 .
321 . * --- 3.3 Vzdělání (Hybridní) ---
322 . * Používáme předpřipravenou 'education_hybrid' proměnnou
323 . * Normalizované hodnoty: "bachelor", "master", "highschool", "missing"
324 . replace education_hybrid = "highschool" if education_hybrid == "high school"
    (2 real changes made)

325 . encode education_hybrid, generate(edu_cat)

326 . label variable edu_cat "Pozadovane vzdelani (kategoricke)"

327 .
328 . * --- 3.4 Zkušenosti ---
329 . * experience_min_llm by měl být již numerický, ale pro jistotu
330 . destring experience_min_llm, replace force
    experience_min_llm already numeric; no replace

331 . label variable experience_min_llm "Min. pozadovane roky zkusenosti"

332 .
333 . * Kategorie zkušeností
334 . gen exp_category = .
    (17,857 missing values generated)

335 . replace exp_category = 1 if experience_min_llm == 0 | experience_min_llm == .
    (3,348 real changes made)

336 . replace exp_category = 2 if experience_min_llm > 0 & experience_min_llm <= 2
    (2,460 real changes made)

337 . replace exp_category = 3 if experience_min_llm > 2 & experience_min_llm <= 5
    (8,411 real changes made)

338 . replace exp_category = 4 if experience_min_llm > 5 & experience_min_llm <= 10
    (3,419 real changes made)

339 . replace exp_category = 5 if experience_min_llm > 10 & experience_min_llm < .
    (219 real changes made)

```

```

340 .
341 . label define exp_lbl 1 "Entry (0)" 2 "Junior (1-2)" 3 "Mid (3-5)" 4 "Senior (6-10)" 5 "Expert"
342 . label values exp_category exp_lbl
343 . label variable exp_category "Kategorie seniority"

344 .
345 . * --- 3.5 Plat ---
346 . destring salary_min salary_mid salary_max, replace force
    salary_min already numeric; no replace
    salary_mid already numeric; no replace
    salary_max already numeric; no replace

347 . replace salary_mid = . if salary_mid < 20000 | salary_mid > 500000
    (908 real changes made, 908 to missing)

348 . label variable salary_mid "Rocni plat - stredni hodnota (USD)"

349 .
350 . * --- 3.6 Sektor a Lokace ---
351 . replace sector = "Unknown" if sector == ""
    (3,504 real changes made)

352 . replace industry = "Unknown" if industry == ""
    (3,505 real changes made)

353 . encode sector, generate(sector_num)

354 .
355 . replace state = "Unknown" if state == ""
    (4,549 real changes made)

356 . encode state, generate(state_num)

357 .
358 . * 3.7 Remote práce ---
359 . * Zde odstranime "capture confirm" abychom videli jestli promenna existuje
360 . gen is_remote = 0

361 . replace is_remote = 1 if strpos(lower(remote_work_types), "home") > 0
    (5,070 real changes made)

362 . replace is_remote = 1 if strpos(lower(remote_work_types), "remote") > 0
    (0 real changes made)

363 .
364 . label variable is_remote "Moznost remote prace (1=ano)"

365 .
366 .
367 .
368 . * =====
369 . * 4. DESKRIPTIVNÍ STATISTIKA
370 . * =====
371 . * Základní přehled datasetu - toto jde typicky do první tabulky v diplomce
372 .
373 . display _n "=====
```

```
=====
```

374 . display "4. DESKRIPTIVNI STATISTIKA"

4. DESKRIPTIVNI STATISTIKA

375 . display "=====

376 .
 377 . * --- 4.1 Frekvence AI tier klasifikace ---
 378 . * KLÍČOVÁ TABULKA: Kolik % pozic vyžaduje AI dovednosti?
 379 . display _n "--- 4.1 Distribuce AI pozadavku v IT pozicich ---"

--- 4.1 Distribuce AI pozadavku v IT pozicich ---

380 . tab desc_tier_llm, missing

desc_tier_llm	Freq.	Percent	Cum.
ai_integration	2,354	13.18	13.18
applied_ai	1,307	7.32	20.50
core_ai	6	0.03	20.54
none	14,190	79.46	100.00
Total	17,857	100.00	

381 . tab desc_tier_llm if desc_tier_llm != "missing", sort

desc_tier_llm	Freq.	Percent	Cum.
none	14,190	79.46	79.46
ai_integration	2,354	13.18	92.65
applied_ai	1,307	7.32	99.97
core_ai	6	0.03	100.00
Total	17,857	100.00	

382 .
 383 . * Procentuální rozdělení (pro text diplomky)
 384 . count if has_ai == 1
3,477

385 . local n_ai = r(N)

386 . count
17,857

387 . local n_total = r(N)

388 . display _n "Podil pozic s AI pozadavky: " %5.2f (`n\_ai'/`n_total')*100 "%"

Podil pozic s AI pozadavky: 19.47%

389 .
 390 . * --- 4.2 Požadované vzdělání ---
 391 . display _n "--- 4.2 Distribuce vzdelavacich pozadavku ---"

--- 4.2 Distribuce vzdelavacich pozadavku ---

392 . tab edu_cat, missing

Pozadovane vzdelani (kategorick e)	Freq.	Percent	Cum.
associate	356	1.99	1.99
bachelor	10,357	58.00	59.99
highschool	411	2.30	62.29
master	352	1.97	64.27
missing	6,368	35.66	99.93
phd	13	0.07	100.00
Total	17,857	100.00	

```
393 . tab edu_cat if edu_cat > 0
```

Pozadovane vzdelani (kategorick e)	Freq.	Percent	Cum.
associate	356	1.99	1.99
bachelor	10,357	58.00	59.99
highschool	411	2.30	62.29
master	352	1.97	64.27
missing	6,368	35.66	99.93
phd	13	0.07	100.00
Total	17,857	100.00	

```
394 .
```

```
395 . * Porovnání vzdělání podle AI tier
```

```
396 . display_n "Vzdelani x AI tier (kontingencni tabulka):"
```

Vzdelani x AI tier (kontingencni tabulka):

```
397 . tab edu_cat ai_tier_num if edu_cat > 0, chi2 column
```

Key
<i>frequency</i> <i>column percentage</i>

Pozadovane vzdelani (kategoric ke)	ai_tier_num				Total
	ai_integr	applied_a	core_ai	none	
associate	20 0.85	9 0.69	0 0.00	327 2.30	356 1.99
bachelor	1,214 51.57	702 53.71	1 16.67	8,440 59.48	10,357 58.00
highschool	28 1.19	20 1.53	0 0.00	363 2.56	411 2.30
master	26 1.10	39 2.98	2 33.33	285 2.01	352 1.97
missing	1,066 45.28	532 40.70	2 33.33	4,768 33.60	6,368 35.66
phd	0 0.00	5 0.38	1 16.67	7 0.05	13 0.07
Total	2,354 100.00	1,307 100.00	6 100.00	14,190 100.00	17,857 100.00

Pearson chi2(15) = 461.9818 Pr = 0.000

```

398 .
399 . * --- 4.3 Požadované zkušenosti ---
400 . display _n "--- 4.3 Distribuce požadavku na zkušenosti ---"

```

--- 4.3 Distribuce požadavku na zkušenosti ---

```
401 . summarize experience_min_llm, detail
```

Min. požadované roky zkušenosti

Percentiles		Smallest		
1%	0	0		
5%	.5	0		
10%	1	0	Obs	15,262
25%	3	0	Sum of Wgt.	15,262
50%	5		Mean	4.520393
		Largest	Std. Dev.	2.62528
75%	5	20		
90%	8	25	Variance	6.892093
95%	10	25	Skewness	.9727604
99%	12	35	Kurtosis	6.38147

```

402 .
403 . tab exp_category, missing

```

Kategorie seniority	Freq.	Percent	Cum.
Entry (0)	3,348	18.75	18.75
Junior (1-2)	2,460	13.78	32.53
Mid (3-5)	8,411	47.10	79.63
Senior (6-10)	3,419	19.15	98.77
Expert (10+)	219	1.23	100.00
Total	17,857	100.00	

```
404 . tab exp_category ai_tier_num, chi2 column
```

Key
<i>frequency</i> <i>column percentage</i>

Kategorie seniority	ai_tier_num				Total
	ai_integr	applied_a	core_ai	none	
Entry (0)	341 14.49	222 16.99	1 16.67	2,784 19.62	3,348 18.75
Junior (1-2)	276 11.72	153 11.71	1 16.67	2,030 14.31	2,460 13.78
Mid (3-5)	1,238 52.59	618 47.28	2 33.33	6,553 46.18	8,411 47.10
Senior (6-10)	472 20.05	289 22.11	2 33.33	2,656 18.72	3,419 19.15
Expert (10+)	27 1.15	25 1.91	0 0.00	167 1.18	219 1.23
Total	2,354 100.00	1,307 100.00	6 100.00	14,190 100.00	17,857 100.00

Pearson chi2(12) = 77.2871 Pr = 0.000

```

405 .
406 . * --- 4.4 Platy ---
407 . display _n "---- 4.4 Distribuce platu ----"

```

--- 4.4 Distribuce platu ---

```
408 . summarize salary_mid, detail
```

Rocni plat - stredni hodnota (USD)

Percentiles		Smallest		
1%	56538	20000		
5%	70937	30000		
10%	79863	30000	Obs	13,743
25%	95000	32640	Sum of Wgt.	13,743
50%	118480		Mean	125240.4
		Largest	Std. Dev.	42210
75%	148750	445000		
90%	180000	445000	Variance	1.78e+09
95%	200000	445000	Skewness	1.370318
99%	245000	450000	Kurtosis	7.794587

```

409 .
410 . * Platy podle AI tier - DŮLEŽITĚ pro argument o "AI premium"
411 . display _n "Plat podle AI tier:"

```

Plat podle AI tier:

```
412 . tabstat salary_mid, by(desc_tier_llm) statistics(count mean sd min p25 p50 p75 max)
```

Summary for variables: salary_mid
by categories of: desc_tier_llm

desc_tier_llm	N	mean	sd	min	p25	p50	p75	max
ai_integration	1869	141477.4	46393.67	36500	107987	135000	169600	410000
applied_ai	1041	151004.5	48527	46532	116373	149500	180000	445000
core_ai	4	168045.8	55785.28	100558	129279	167937.5	206812.5	235750
none	10829	119945.5	39048.28	20000	92500	114067	140000	450000
Total	13743	125240.4	42210	20000	95000	118480	148750	450000

```

413 .
414 . * --- 4.5 Sektory a industrie ---
415 . display _n "---- 4.5 Top 10 sektoru ----"

```

--- 4.5 Top 10 sektoru ---

```
416 . tab sector if sector != "Unknown", sort
```

sector	Freq.	Percent	Cum.
Information Technology	6,074	42.32	42.32
Manufacturing	1,219	8.49	50.81
Financial Services	1,081	7.53	58.34
Aerospace & Defense	981	6.83	65.18
Management & Consulting	707	4.93	70.10
Healthcare	692	4.82	74.93
Media & Communication	546	3.80	78.73
Retail & Wholesale	405	2.82	81.55
Education	375	2.61	84.16
Insurance	321	2.24	86.40
Human Resources & Staffing	294	2.05	88.45
Government & Public Administration	271	1.89	90.34
Pharmaceutical & Biotechnology	229	1.60	91.93
Energy, Mining & Utilities	184	1.28	93.21
Telecommunications	175	1.22	94.43
Construction, Repair & Maintenance Se..	171	1.19	95.62
Transportation & Logistics	143	1.00	96.62
Arts, Entertainment & Recreation	103	0.72	97.34

Real Estate	94	0.65	97.99
Nonprofit & NGO	76	0.53	98.52
Personal Consumer Services	56	0.39	98.91
Restaurants & Food Service	48	0.33	99.25
Legal	44	0.31	99.55
Hotels & Travel Accommodation	33	0.23	99.78
Agriculture	30	0.21	99.99
Media	1	0.01	100.00
Total	14,353	100.00	

417 .
 418 . display _n "--- 4.6 Remote prace ---"

--- 4.6 Remote prace ---

419 . tab is_remote

Možnost remote prace (1=ano)	Freq.	Percent	Cum.
0	12,787	71.61	71.61
1	5,070	28.39	100.00
Total	17,857	100.00	

420 . tab is_remote has_ai, chi2 row

Key
<i>frequency</i> <i>row percentage</i>

Možnost remote prace (1=ano)	has_ai		Total
	0	1	
0	10,618 83.04	2,169 16.96	12,787 100.00
1	3,762 74.20	1,308 25.80	5,070 100.00
Total	14,380 80.53	3,477 19.47	17,857 100.00

Pearson chi2(1) = 180.7841 Pr = 0.000

421 .
 422 .
 423 . * =====
 424 . * 5. ANALYTICKÉ TESTY - HYPOTÉZY
 425 . * =====
 426 . * Statistické testy pro ověření hypotéz diplomové práce
 427 .

```
428 . display _n "=====
```

```
=====
```

```
429 . display "5. STATISTICKE TESTY"
5. STATISTICKE TESTY
```

```
430 . display "=====
```

```
=====
```

```
431 .
432 . * --- 5.1 T-test: Liší se platy AI vs non-AI pozic? ---
433 . * H0: Průměrný plat AI pozic = průměrný plat non-AI pozic
434 . * H1: Průměrné platy se liší
435 .
436 . display _n "--- 5.1 T-test: Plat AI vs non-AI pozic ---"
```

```
--- 5.1 T-test: Plat AI vs non-AI pozic ---
```

```
437 . ttest salary_mid, by(has_ai)
```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
0	10,988	120345.9	375.8993	39403.15	119609.1	121082.8
1	2,755	144761.4	897.5534	47110.87	143001.5	146521.4
combined	13,743	125240.4	360.0597	42210	124534.6	125946.2
diff		-24415.49	874.948		-26130.5	-22700.47

```
diff = mean(0) - mean(1) t = -27.9051
Ho: diff = 0 degrees of freedom = 13741
```

```
Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = 0.0000 Pr(|T| > |t|) = 0.0000 Pr(T > t) = 1.0000
```

```
438 .
439 . * Efekt size (Cohenovo d) - důležité pro interpretaci praktické významnosti
440 . quietly summarize salary_mid if has_ai == 0
```

```
441 . local mean_no_ai = r(mean)
```

```
442 . local sd_no_ai = r(sd)
```

```
443 . local n_no_ai = r(N)
```

```
444 .
445 . quietly summarize salary_mid if has_ai == 1
```

```
446 . local mean_ai = r(mean)
```

```
447 . local sd_ai = r(sd)
```

```
448 . local n_ai = r(N)
```

```
449 .
450 . * Pooled standard deviation
```

```
451 . local sd_pooled = sqrt(((`n_no_ai'-1)*`sd_no_ai'^2 + (`n_ai'-1)*`sd_ai'^2) / (`n_no_ai'+`n_ai'
```

```
452 . local cohens_d = (`mean_ai' - `mean_no_ai') / `sd_pooled'
```

```
453 .
```

```
454 . display _n "Cohenovo d (effect size): " %5.3f `cohens_d'
```

Cohenovo d (effect size): 0.595

```
455 . display "Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt"
Interpretace: |d| < 0.2 = maly, 0.2-0.8 = stredni, > 0.8 = velky efekt
```

```
456 .
```

```
457 . * --- 5.2 ANOVA: Liší se platy mezi AI tiers? ---
```

```
458 . * Testuje rozdíly mezi none, ai_integration, ai_focused
```

```
459 .
```

```
460 . display _n "--- 5.2 ANOVA: Plat podle AI tier ---"
```

--- 5.2 ANOVA: Plat podle AI tier ---

```
461 . oneway salary_mid ai_tier_num, tabulate bonferroni
```

ai_tier_num	Summary of Rocni plat - stredni hodnota (USD)		
	Mean	Std. Dev.	Freq.
ai_integr	141477.39	46393.668	1,869
applied_a	151004.53	48526.999	1,041
core_ai	168045.75	55785.282	4
none	119945.51	39048.279	10,829
Total	125240.41	42209.997	13,743

Source	Analysis of Variance			F	Prob > F
	SS	df	MS		
Between groups	1.4947e+12	3	4.9823e+11	297.75	0.0000
Within groups	2.2989e+13	13739	1.6733e+09		
Total	2.4484e+13	13742	1.7817e+09		

Bartlett's test for equal variances: $\chi^2(3) = 172.4292$ Prob> $\chi^2 = 0.000$

Comparison of Rocni plat - stredni hodnota (USD) by ai_tier_num
(Bonferroni)

Row Mean- Col Mean	ai_integ	applied_	core_ai
applied_	9527.14 0.000		
core_ai	26568.4 1.000	17041.2 1.000	
none	-21531.9 0.000	-31059 0.000	-48100.2 0.112

```
462 .
```

```
463 . * --- 5.3 Chi-square: Vzdělání a AI požadavky ---
```

```
464 . * Jsou AI pozice náročnější na vzdělání?
```

```
465 .
```

```
466 . display _n "--- 5.3 Chi-square: Vzdelani x AI tier ---"
```

--- 5.3 Chi-square: Vzdelani x AI tier ---

```
467 . tab edu_cat has_ai if edu_cat > 0, chi2 expected
```

Key
<i>frequency</i>
<i>expected frequency</i>

Pozadovane vzdelani (kategoric ke)	has_ai		Total
	0	1	
associate	328 286.7	28 69.3	356 356.0
bachelor	8,543 8,340.4	1,814 2,016.6	10,357 10,357.0
highschool	365 331.0	46 80.0	411 411.0
master	286 283.5	66 68.5	352 352.0
missing	4,851 5,128.1	1,517 1,239.9	6,368 6,368.0
phd	7 10.5	6 2.5	13 13.0
Total	14,380 14,380.0	3,477 3,477.0	17,857 17,857.0

Pearson chi2(5) = 156.7363 Pr = 0.000

```
468 .
469 . * --- 5.4 Chi-square: Zkušnosti a AI požadavky ---
470 . display _n "--- 5.4 Chi-square: Zkusenosti x AI tier ---"
```

--- 5.4 Chi-square: Zkusenosti x AI tier ---

```
471 . tab exp_category has_ai, chi2 expected
```

Key
<i>frequency</i>
<i>expected frequency</i>

Kategorie seniority	has_ai		Total
	0	1	
Entry (0)	2,809 2,696.1	539 651.9	3,348 3,348.0
Junior (1-2)	2,052 1,981.0	408 479.0	2,460 2,460.0
Mid (3-5)	6,655 6,773.3	1,756 1,637.7	8,411 8,411.0
Senior (6-10)	2,693 2,753.3	726 665.7	3,419 3,419.0
Expert (10+)	171 176.4	48 42.6	219 219.0
Total	14,380 14,380.0	3,477 3,477.0	17,857 17,857.0

Pearson chi2(4) = 55.5656 Pr = 0.000

```
472 .
473 . * --- 5.5 Mann-Whitney U test (neparametrický) ---
474 . * Pro případ, že platy nemají normální rozdělení
475 .
476 . display _n "--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---"
```

--- 5.5 Mann-Whitney U test: Plat AI vs non-AI ---

```
477 . ranksum salary_mid, by(has_ai)
```

Two-sample Wilcoxon rank-sum (Mann-Whitney) test

has_ai	obs	rank sum	expected
0	10988	70557704	75509536
1	2755	23884192	18932360
combined	13743	94441896	94441896

unadjusted variance 3.467e+10

adjustment for ties -595872.38

adjusted variance 3.467e+10

Ho: salary~d(has_ai==0) = salary~d(has_ai==1)

z = -26.594

Prob > |z| = 0.0000

```
478 .
479 .
480 . * =====
481 . * 6. REGRESNÍ ANALÝZA
482 . * =====
483 . * Modelování faktorů ovlivňujících plat a AI požadavky
484 .
485 . display _n "=====
```

=====

```
486 . display "6. REGRESNI ANALYZA"
487 . display "=====
```

=====

```
488 .
489 . * --- 6.1 OLS regrese: Co ovlivňuje plat? ---
490 . * Závislá proměnná: salary_mid
491 . * Nezávislé: has_ai, edu_level, experience_min_llm, is_remote
492 .
493 . display _n "--- 6.1 OLS regrese: Determinanty platu ---"
```

--- 6.1 OLS regrese: Determinanty platu ---

```
494 .
495 . regress salary_mid has_ai i.edu_cat experience_min_llm is_remote
```

Source	SS	df	MS	Number of obs	=	11,899
Model	4.3534e+12	8	5.4417e+11	F(8, 11890)	=	402.51
Residual	1.6075e+13	11,890	1.3520e+09	Prob > F	=	0.0000
				R-squared	=	0.2131
				Adj R-squared	=	0.2126
Total	2.0428e+13	11,898	1.7169e+09	Root MSE	=	36769

salary_mid	Coef.	Robust Std. Err.	t	P> t	[95% Conf. Interval]	
has_ai	19410.84	8892.01	2.18	0.029	1981.044	36840.63
edu_cat						
bachelor	12236.5	2020.524	6.06	0.000	8275.942	16197.06
highschool	7831.508	2607.258	3.00	0.003	2720.855	12942.16
master	17074.39	2952.829	5.78	0.000	11286.36	22862.42
missing	16571.38	2153.314	7.70	0.000	12350.53	20792.23
phd	34644.42	7163.876	4.84	0.000	20602.05	48686.79
edu_cat#c.has_ai						
bachelor	-2364.461	8962.538	-0.26	0.792	-19932.5	15203.58
highschool	-14773.18	10217.54	-1.45	0.148	-34801.24	5254.876
master	8144.445	11337.29	0.72	0.473	-14078.49	30367.38
missing	7096.398	9060.349	0.78	0.434	-10663.37	24856.16
phd	8178.067	30108.66	0.27	0.786	-50839.83	67195.97
experience_min_llm	5458.089	158.7991	34.37	0.000	5146.817	5769.362
is_remote	11639.64	853.8972	13.63	0.000	9965.857	13313.41
_cons	81045.14	1997.531	40.57	0.000	77129.65	84960.62

```

506 .
507 . * --- 6.3 Logistická regrese: Co predikuje AI požadavky? ---
508 . * Které charakteristiky pozice souvisí s AI požadavky?
509 .
510 . display _n "--- 6.3 Logisticka regrese: Prediktory AI pozadavku ---"

```

--- 6.3 Logisticka regrese: Prediktory AI pozadavku ---

```

511 . logit has_ai i.edu_cat experience_min_llm is_remote, or

```

```

Iteration 0:  log likelihood = -7640.7448
Iteration 1:  log likelihood = -7497.5094
Iteration 2:  log likelihood = -7494.1649
Iteration 3:  log likelihood = -7494.1444
Iteration 4:  log likelihood = -7494.1444

```

Logistic regression	Number of obs	=	15,262
	LR chi2(7)	=	293.20
	Prob > chi2	=	0.0000
Log likelihood = -7494.1444	Pseudo R2	=	0.0192

has_ai	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
edu_cat						
bachelor	2.417142	.4997532	4.27	0.000	1.611807	3.62486
highschool	1.42392	.3719776	1.35	0.176	.8533399	2.376015
master	2.956445	.747325	4.29	0.000	1.801377	4.85216
missing	3.63699	.755776	6.21	0.000	2.42025	5.465426
phd	8.707554	6.134277	3.07	0.002	2.18899	34.63767
experience_min_llm	1.018785	.0079704	2.38	0.017	1.003283	1.034527
is_remote	1.516053	.0662321	9.52	0.000	1.391643	1.651585
_cons	.073278	.0151196	-12.67	0.000	.0489039	.1098005

Note: **_cons** estimates baseline odds.

```

512 .
513 . * Marginální efekty (interpretovatelné jako procentní změny)
514 . display _n "--- 6.3b Marginalni efekty ---"

```

--- 6.3b Marginalni efekty ---

```
515 . margins, dydx(*) atmeans
```

```

Conditional marginal effects      Number of obs      =      15,262
Model VCE      : OIM

```

```

Expression      : Pr(has_ai), predict()
dy/dx w.r.t.    : 2.edu_cat 3.edu_cat 4.edu_cat 5.edu_cat 6.edu_cat experience_min_llm is_remote
at              : 1.edu_cat      =      .0215568 (mean)
                  2.edu_cat      =      .6126982 (mean)
                  3.edu_cat      =      .025095 (mean)
                  4.edu_cat      =      .0189359 (mean)
                  5.edu_cat      =      .3211244 (mean)
                  6.edu_cat      =      .0005897 (mean)
                  experience~m    =      4.520393 (mean)
                  is_remote       =      .2914428 (mean)

```

	Delta-method				[95% Conf. Interval]	
	dy/dx	Std. Err.	z	P> z		
edu_cat						
bachelor	.0960943	.0160304	5.99	0.000	.0646754	.1275132
highschool	.0310228	.0225205	1.38	0.168	-.0131166	.0751621
master	.1275784	.0290966	4.38	0.000	.0705501	.1846067
missing	.1640223	.0167551	9.79	0.000	.131183	.1968616
phd	.3567673	.1666906	2.14	0.032	.0300596	.6834749
experience_min_llm	.002914	.0012244	2.38	0.017	.0005142	.0053138
is_remote	.0651517	.0068062	9.57	0.000	.0518117	.0784916

Note: dy/dx for factor levels is the discrete change from the base level.

```

516 .
517 .
518 . * =====
519 . * 7. ANALÝZA HARD SKILLS
520 . * =====
521 . * Rozbor nejčastějších technických dovedností
522 .
523 . display _n "=====

```

```
=====
```

```

524 . display "7. ANALYZA HARD SKILLS"
7. ANALYZA HARD SKILLS

```

```

525 . display "=====
=====

```

```

526 .
527 . * Poznámka: hardskills je textový sloupec s čárkami oddělenými skills
528 . * Pro detailní analýzu je lepší použít Python k vytvoření dummy proměnných
529 . * Zde ukážeme základní exploraci
530 .

```



```

531 . * Počet unique skills na pozici (aproximace)
532 . gen skill_count = 1 + length(hardskills) - length(subinstr(hardskills, ",", "", .))

533 . replace skill_count = 0 if hardskills == ""
    (110 real changes made)

534 . label variable skill_count "Pocet pozadovanych hard skills"

535 .
536 . display _n "--- 7.1 Pocet skills na pozici ---"

    --- 7.1 Pocet skills na pozici ---

537 . summarize skill_count, detail

```

Pocet pozadovanych hard skills				
	Percentiles	Smallest		
1%	1	0		
5%	3	0		
10%	5	0	Obs	17,857
25%	9	0	Sum of Wgt.	17,857
50%	15		Mean	16.78658
		Largest	Std. Dev.	10.30584
75%	23	72		
90%	31	72	Variance	106.2103
95%	36	80	Skewness	.9649615
99%	47	80	Kurtosis	4.325341

```

538 . tabstat skill_count, by(desc_tier_llm) statistics(count mean sd min max)

```

Summary for variables: skill_count
by categories of: desc_tier_llm

desc_tier_llm	N	mean	sd	min	max
ai_integration	2354	19.46432	10.3817	0	68
applied_ai	1307	20.62663	10.79929	1	80
core_ai	6	11.16667	6.431692	1	20
none	14190	15.99105	10.09171	0	80
Total	17857	16.78658	10.30584	0	80

```

539 .
540 . * T-test: Vyžadují AI pozice více skills?
541 . display _n "--- 7.2 T-test: Pocet skills AI vs non-AI ---"

```

--- 7.2 T-test: Pocet skills AI vs non-AI ---

```

542 . ttest skill_count, by(has_ai)

```

Two-sample t test with equal variances

Group	Obs	Mean	Std. Err.	Std. Dev.	[95% Conf. Interval]	
0	14,380	16.04652	.0842893	10.10769	15.88131	16.21174
1	3,477	19.84728	.1789442	10.55164	19.49644	20.19813
combined	17,857	16.78658	.0771222	10.30584	16.63542	16.93775
diff		-3.800759	.19268		-4.178431	-3.423088

diff = mean(0) - mean(1) t = **-19.7258**
Ho: diff = 0 degrees of freedom = **17855**

Ha: diff < 0 Ha: diff != 0 Ha: diff > 0
Pr(T < t) = **0.0000** Pr(|T| > |t|) = **0.0000** Pr(T > t) = **1.0000**

```

543 .
544 .
545 . * =====
546 . * 8. EXPORTY PRO TABULKY A GRAFY
547 . * =====
548 . * Příprava dat pro publikovatelné tabulky
549 .
550 . display _n "=====
```

```
=====
```

```

551 . display "8. EXPORTY"
      8. EXPORTY
```

```

552 . display "=====
```

```
=====
```

```

553 .
554 . * --- 8.1 Summary statistics ---
555 . display _n "--- 8.1 Summary statistics pro tabulky ---"
```

```
--- 8.1 Summary statistics pro tabulky ---
```

```
556 . summarize salary_mid experience_min_llm edu_cat skill_count has_ai is_remote
```

Variable	Obs	Mean	Std. Dev.	Min	Max
salary_mid	13,743	125240.4	42210	20000	450000
experience~m	15,262	4.520393	2.62528	0	35
edu_cat	17,857	3.115249	1.448891	1	6
skill_count	17,857	16.78658	10.30584	0	80
has_ai	17,857	.1947136	.3959911	0	1
is_remote	17,857	.2839223	.4509122	0	1

```

557 .
558 . * Export do CSV pro další zpracování (grafy v Excelu/R/Python)
559 . export delimited desc_tier_llm salary_mid experience_min_llm edulevel_llm state sector using
      file ./output/summary_for_charts.csv saved
```

```

560 .
561 .
562 . * =====
563 . * 9. VIZUALIZACE
564 . * =====
565 . * Základní grafy pro diplomovou práci
566 .
567 . display _n "=====
```

```
=====
```

```

568 . display "9. VIZUALIZACE"
      9. VIZUALIZACE
```

```

569 . display "=====
```

```
=====
```

```

570 .
571 . * --- 9.1 Sloupcový graf: AI tier distribuce ---
572 . graph bar (count), over(desc_tier_llm) ///
>     title("Distribuce AI požadavku v IT pozicích") ///
>     ytitle("Pocet pozic") ///
>     bar(1, color(navy))
```

```

573 . graph export "$outdir/ai_tier_distribution.png", replace width(1200)
    (file ./output/ai_tier_distribution.png written in PNG format)

574 .
575 . * --- 9.2 Box plot: Platy podle AI tier ---
576 . graph box salary_mid, over(desc_tier_llm) ///
    > title("Rozložení platu podle AI požadavku") ///
    > ytitle("Rocni plat (USD)")

577 . graph export "$outdir/salary_by_ai_tier.png", replace width(1200)
    (file ./output/salary_by_ai_tier.png written in PNG format)

578 .
579 . * --- 9.3 Histogram: Požadované zkušenosti ---
580 . histogram experience_min_llm if experience_min_llm < 15, ///
    > by(has_ai, title("Požadovane zkusenosti") note("")) ///
    > xtitle("Roky zkusenosti") percent bin(15)

581 . graph export "$outdir/experience_histogram.png", replace width(1200)
    (file ./output/experience_histogram.png written in PNG format)

582 .
583 . * --- 9.4 Vzdělání podle AI tier ---
584 . graph bar (count), over(edu_cat) over(has_ai) ///
    > title("Vzdelavaci pozadavky: AI vs non-AI pozice") ///
    > ytitle("Pocet pozic") ///
    > legend(label(1 "non-AI") label(2 "AI pozice"))

585 . graph export "$outdir/education_by_ai.png", replace width(1200)
    (file ./output/education_by_ai.png written in PNG format)

586 .
587 .
588 . * =====
589 . * 10. ZÁVĚR A ULOŽENÍ
590 . * =====
591 .
592 . display _n "=====
    =====

593 . display "ANALYZA DOKONCENA"
ANALYZA DOKONCENA

594 . display "=====
    =====

595 . display "Vystupy ulozeny do: $outdir"
Vystupy ulozeny do: ./output

596 . display "Log soubor: $outdir/ai_skills_analysis_`time_string`.log"
Log soubor: ./output/ai_skills_analysis_24_Jan_2026_12-27-12.log

597 .
598 . * Ulož zpracovaný dataset pro další práci
599 . save "$outdir/ai_skills_processed.dta", replace
    file ./output/ai_skills_processed.dta saved

600 .
601 . * Zavři log

```

```
602 . log close
      name: <unnamed>
      log: C:\Users\murcj\Projects\glassdoor-jobs-ai-skills\analysis\stata\output\ai_skills_a
      log type: text
      closed on: 24 Jan 2026, 12:29:11
```

```
603 .
604 . * =====
605 . * KONEC DO-FILU
606 . * =====
607 .
      end of do-file

608 .
```